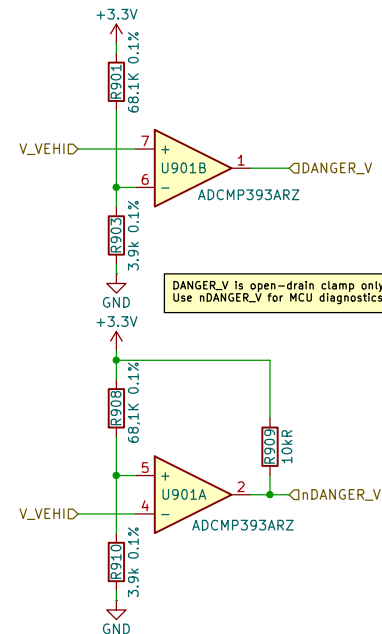
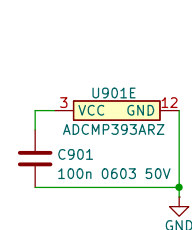
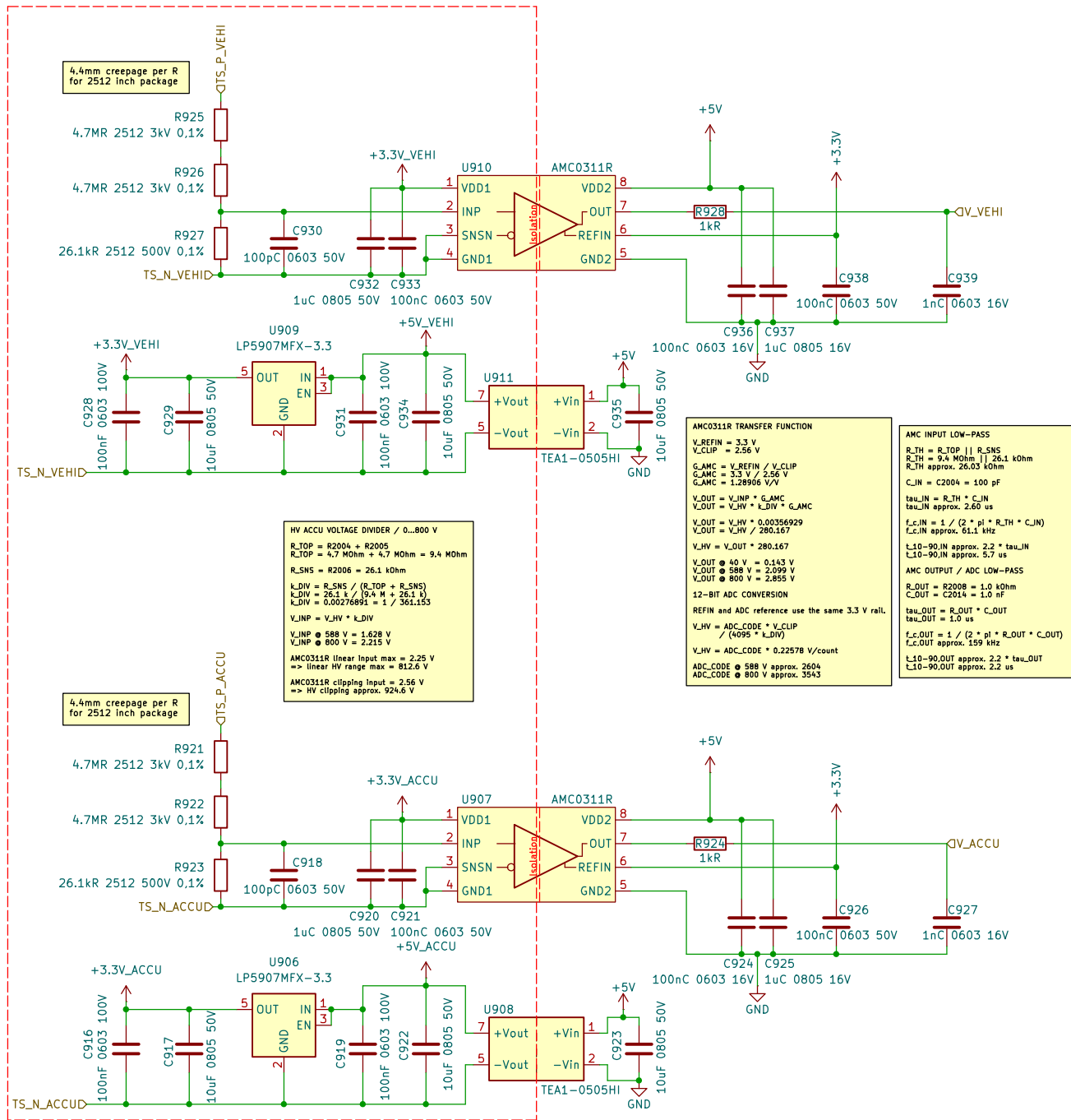




DANGER VOLTAGE THRESHOLD: APPROX. 50 V

HV measurement transfer:
 $R_{TOP} = 4.7\text{ MOhm} + 4.7\text{ MOhm} = 9.4\text{ MOhm}$
 $R_{SNS} = 26.1\text{ kOhm}$
 $k_{DIV} = 26.1\text{ k} / (9.4\text{ M} + 26.1\text{ k})$
 $k_{DIV} = 0.00276891$

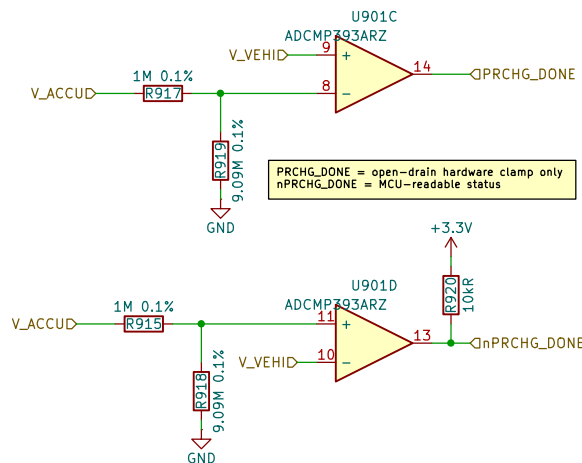
AMC0311R:
 $V_{OUT} = V_{HV} * k_{DIV} * V_{REFIN} / 2.56\text{ V}$
 With $V_{REFIN} = 3.3\text{ V}$:
 $V_{OUT} = V_{HV} * 0.00356930$
 $V_{OUT} @ 50\text{ V} = 178.46\text{ mV}$

Comparator threshold reference:
 $R_{REF_HIGH} = 68.1\text{ kOhm}$
 $R_{REF_LOW} = 3.90\text{ kOhm}$
 $V_{DANGER_REF} = 3.3\text{ V} * 3.90\text{ k} / (68.1\text{ k} + 3.90\text{ k})$
 $V_{DANGER_REF} = 178.75\text{ mV}$

Nominal HV threshold:
 $V_{HV,TH} = 178.75\text{ mV} / 0.00356930$
 $V_{HV,TH} = 50.08\text{ V}$

DANGER_V:
 $IN+ = V_{VEHI}$
 $IN- = V_{DANGER_REF}$
 Output released when $V_{HV} > \text{approx. } 50\text{ V}$

nDANGER_V:
 $IN+ = V_{DANGER_REF}$
 $IN- = V_{VEHI}$
 Output LOW when $V_{HV} > \text{approx. } 50\text{ V}$



Precharge 90% check:
 V_{ACCUD} and V_{VEHI} are equally scaled HV measurements.
 $V_{ACC_90} = V_{ACCUD} * R_{LOW} / (R_{HIGH} + R_{LOW})$
 $R_{HIGH} = 1.00\text{M}$
 $R_{LOW} = 9.09\text{M}$
 $V_{ACC_90} \approx 0.900 * V_{ACCUD}$

PRCHG_DONE:
 $IN+ = V_{VEHI}$
 $IN- = V_{ACC_90}$
 HIGH when $V_{VEHI} >= 0.9 * V_{ACCUD}$
 Used as MCU status input.

AIR_P interlock:
 Same comparator condition is also used as open-drain clamp without pullup at AIR_P driver.
 Comparator can only block AIR_P, not enable it.
 Firmware should close AIR_P only after $V_{VEHI} >= 95\% * V_{ACCUD}$ with debounce.
 Hardware lockout threshold: 90%

Lukas Elektronik

Sheet: /HV_Measurement/
 File: HV_Measurement.kicad_sch

Title: PackController

Size: A3 Date: 2026-03-03
 KiCad E.D.A. 10.0.3

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